Slambient Master Blueprint: The 4-Layer Hybrid Ambient Workstation (Complete Production Edition)

This master architecture document locks in all finalized design elements, polyphony budgets, generative logic, synesthetic color-mapping UI themes, UI layout specifications, production-level reliability features (state persistence, brickwall limiting, MIDI clock sync), preset management, and plug-and-play instant generative modes for the custom **Slambient** (Sloop-FM1) firmware project.

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1. Aesthetic, Sonic Goals & Maestro Inspirations

- **The Vibe:** Warm, nostalgic, cinematic "Boards of Canada" ambient aesthetic with organic drift, emotional depth, and zero digital popping/clipping.
- **Analog Imperfections:** Built-in tape flutter / wow-and-flutter modulation running across the master output chain.
- **Dynamic Breathing & Dynamic Ducking:** Designed with natural pauses, velvet-smooth attacks, and soft rolling envelopes so the "wall of sound" blooms and rests organically. Subtle dynamic ducking (-3 dB to -6 dB) is automatically applied to the sub-drone and shimmer arpeggio during active lead phrases, maximizing harmonic clarity and space without adding extra voice burden.
- **Brian Eno Engine (Generative Phase-Locked Ecosystems & Coprime Architecture):**
    - **Coprime Multi-Layer Clock & Clock Dividers:** Multi-layer step cycles utilize extended coprime lengths (7, 11, 13, 17, 19, and 23 steps) combined with layer clock dividers (e.g., Sub-Drone steps advance every 4 bars / 64 steps, Chords advance every 2 bars / 32 steps). While raw 7-, 11-, and 13-step coprime cycles repeat every 1,001 steps (~8 minutes at ambient tempos), combining clock dividers with extended coprimes pushes structural loop periods to many hours.
    - **True Non-Repetition, Weighted Random Walk & "Weather" Macro Layer:** Ultimate non-repetition is guaranteed by a weighted random walk with tonic/dominant gravitational pull (where interval probabilities naturally bound discordance to a 10% tension rule) and a 20–40 minute ultra-slow drift "Weather" macro layer that modulates global density and harmonic behavior over continuous multi-hour sessions.
- **Hiroshi Yoshimura Engine (Kankyo Ongaku Fluidity):**
    - **Environmental Micro-Glides (Thermal Portamento):** Fluid, variable-speed portamento curves that make harmonic shifts bend and "melt" naturally like water currents rather than stepping abruptly.
    - **Acoustic Spatial Diffusion:** Algorithmic feedback and stereo expansion scaling dynamically with drone spectral weight to simulate physical architectural space.

- **Four Tet Textures:** Acoustic-to-pad hybrid transient envelopes pairing woody plucks with lush washes.
- **Aphex Tuning & Micro-Timing:** Alternative micro-tunings, sub-millisecond event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) for sub-millisecond micro-timing humanization, and micro-timing swing for organic human feel.

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2. The 4-Layer Slambient Architecture

To maximize the FM-1 hardware without cluttering the frequency spectrum, the engine is divided into four distinct, concurrent layers running off a unified master clock:

1. **Layer 1: Euclidean Rhythms (The Pulse)**
    - Generates organic, syncopated rhythmic percussion and micro-accents.
    - Uses ultra-short percussive envelopes that decay instantly so they never steal voices from the pads.
2. **Layer 2: Ambient Chords & Lead Engine (The Harmony & Voice)**
    - **Weighted Random Walk & Tonic Gravity:** Discards rigid loops in favor of a weighted random walk driven by tonic/dominant gravitational pull and interval probability distributions across scale degrees (Dorian/Lydian modal shifts), anchoring pitch selection to tonal centers.
    - **Call-and-Response Interplay:** Meandering lead phrases trigger *first* with clear presence, followed by a breathing pause and a responsive pad swell anchored to that musical space.
    - **Controlled Discordance & Natural 10% Tension Rule:** Tightly bounded interval step probabilities ensure that the 10% tension rule emerges naturally from distribution math, resolving harmonically back to root stability without artificial thresholding.
3. **Layer 3: Micro-Arpeggio Generator (The Shimmer)**
    - An independent, fast-moving melodic layer that trickles notes in a randomized, non-repeating high-mid cascade.
    - Shaped with delicate velocities to sound like twinkling mist or falling water over the pad floor.
4. **Layer 4: Living Sub-Drone Bed (The Tectonic Floor)**
    - A continuous, slow-evolving spectral bed that morphs its harmonic balance over minutes rather than seconds.
    - Tracks the active root anchor and slowly swells sub-bass overtones to keep emotional tension grounded.

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3. Plug-and-Play Instant Ambient Mode ("Instant Slambient")

- **Zero-Thought Operation:** A dedicated startup mode allowing you to simply turn on the FM-1, press play, and let the device fully build out a complete, self-sustaining ambient world instantly.
- **Mood Presets:** Pre-baked macro profiles (e.g., *Nostalgic Drift, Glacial Morning, Hauntological Tape*) that automatically configure scale parameters, drone weight, Euclidean density, and arpeggio speed depending on the desired mood.

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4. Polyphony Management, Audio Safety & Memory Constraints

- **Generative Co-Processor Architecture:** Slambient operates as an internal Generative Event/MIDI Co-Processor triggering notes, CCs, and envelopes into the stock SLOOP/Felucca synthesis and drum engine (Tracks 1–4) rather than attempting to synthesize PCM waveforms from scratch inside the high-priority audio ISR. The pre-computed `SINE_LUT[1024]` lookup table is used solely for low-frequency control signals (LFOs), tape wow/flutter delay modulation, thermal portamento pitch curves, and synesthetic UI polygon rendering.
- **DSP Asymmetry Principle (Drum vs. Synth Voice Cost):** Drum voices and synth voices run on completely separate DSP pipelines and are not 1:1 interchangeable in CPU cycles. Drum voices are lightweight one-shot triggers with short decay (50–200 ms), whereas synth voices (especially 6-op FM6 and 4-pole resonant VA) require intensive per-sample DSP calculations.
- **Finalized Polyphony Strategy (12 Managed Voices Total):** The polyphony budget is optimized across DSP workloads to ensure absolute CPU safety and stability:
    - **Cap Drum Voices at 4 (Layer 1):** Capped at 4 drum voices (reduced from stock 6). Euclidean ambient percussion only fires 2–3 simultaneous micro-accents (kick, woody click, shaker), so capping at 4 eliminates unnecessary CPU spikes during dense polyrhythmic passages. Includes global Percussion Filter Damp and transient softening to gently integrate percussive textures.
    - **Guarantee CPU Headroom for Master FX & Lean Half-Rate Reverb:** The reclaimed CPU margin directly protects the master bus, ensuring tape wow/flutter delay modulation, tanh soft-saturation with a short lookahead peak limiter with smooth release placed directly post-saturation to catch any inter-sample overshoots, and deep spatial diffusion reverb run with zero buffer underruns or audio stuttering. The diffusion reverb network runs at half sample rate (22.05 kHz) with a low-pass darker tail, cutting reverb memory requirements in half (~15–20 KB) to fit inside the 344 KB audio heap while perfectly evoking vintage Boards of Canada Lexicon / Space Echo warmth. To prevent hung MIDI voices during complex generative passages, an automatic Stuck-Note Watchdog enforces layer-specific rules: Lead, Arp, and Drums have a 15-second maximum note duration; Chord Pads have a 30-second maximum duration with seamless

50 ms crossfade re-triggering; Sub-Drone (Layer 4) is completely exempt from watchdog timeouts. An all-notes-off panic handler is also available.
    - *Layer 4 (Living Sub-Drone Bed):* 1 synth voice (mono continuous bed).
    - *Layer 2 (Ambient Chords & Lead Engine):* 3 synth voices (open triad voicings). Layer 2 chords and the meandering lead share voices dynamically: when the lead plays, it borrows Voice 3 of the chord triad (thinning the pad to an open 5th / power chord), which then swells back into a full triad during rests.
    - *Layer 3 (Micro-Arpeggio Shimmer):* 1 synth voice (mono cascade).
    - **Explicit Voice-Stealing Hierarchy & Amplitude Envelopes:** Voice preemption follows a strict priority hierarchy: `Live Keys (PRIORITY_CRITICAL)` > `Lead Melodic Voice` > `Chord Pad Voicings` > `Shimmer Arp` > `Sub-Drone (LOWEST)`. Voice transitions are controlled by amplitude envelopes rather than zero-crossing detectors: when a voice is preempted, it executes a 10 ms cosine amplitude ramp-down to eliminate preemption clicks during voice reallocation, while chord and modal transitions use a 50–100 ms exponential amplitude release.
    - **Preserve the Reliable 8-Synth Pool:** Rather than risking RAM overflow or ISR starvation by pushing to 9+ synth voices, keep 8 synth voices and resolve harmonic density via dynamic Voice Borrowing (lead voice borrows Voice 3 of the chord pad during phrases, which swells back into a full triad during breathing pauses).
    - **Final Voice Budget Summary:** 8 Synth + 4 Drum = 12 total managed voices (1 drone, 3 chord pad / lead borrowed, 1 shimmer arp, 3 reserved live keys, 4 Euclidean percussion).
- **Zero-Crossing Audio Safety & Interpolation:** Smooth voice release windows (50–100 ms exponential fade) are enforced during modal shifts and auto-latch generative hand-offs to avoid zero-crossing clicks. Delay-line read pointer modulations for wow/flutter and thermal portamento utilize linear/hermite interpolation to completely prevent zipper noise.
- **Master Brickwall Limiter & Soft-Saturation:** Master bus processing combines tanh soft-saturation with a true hard peak ceiling at -0.5 dBFS placed directly post-saturation to catch inter-sample overshoots, preserving warm tape grit and preventing digital clipping during dense overlapping voice swells.
- **RAM & Display Strip-Rendering Architecture:** Static `.data+.bss` RAM has a strict ~98 KB ceiling with only ~5.5 KB of headroom remaining. Allocating a full 240x240 16-bit display framebuffer (~115 KB) exceeds available RAM, and full-screen 240x240 16-bit rendering over SPI takes 23–46 ms, exceeding the 41 ms (24 FPS) rendering budget. The UI avoids full-screen per-pixel watercolor blending, using instead coarse blended color zones and strictly bounded dirty rectangles (orbital arcs, center text, and dials). The 4–8 KB canvas tile is allocated once at boot from the dynamic audio heap, marked non-cacheable/DMA-capable, with zero runtime allocation or freeing to preserve the 5.5 KB BSS RAM. Sub-millisecond event scheduling handles event offsets within 64-sample audio blocks (~1.45 ms at 44.1 kHz) to preserve micro-timing. Furthermore, all

generative and modulation math uses 32-bit fixed-point integer arithmetic referencing the `SINE_LUT[1024]` table to prevent floating-point pipeline stalls inside audio interrupts.

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5. Hardware Integration & Real-World Limitations

- **Live Pass-Through:** Actively playing the device's 27 silicone keys bypasses generation and sends your notes straight to the synth engine.
- **Leaky Pitch-Class Histogram & Mode Inference:** As manual performance notes are played on the 27 silicone keys, a leaky pitch-class histogram analyzes note distribution in real-time, inferring both the root anchor and scale mode (e.g., Dorian vs. Lydian) from manual play.
- **Auto-Latch Seamless Transition:** When no keys are touched for a brief threshold (e.g., 3 seconds), an internal flag triggers a harmonic hand-off. The engine captures the inferred root and modal scale, seamlessly reassigning them as the new generative anchor for wandering bass, lead, and pads.
- **Bluetooth BLE Hardware Disable:** In custom SLOOP/Felucca firmwares, the Bluetooth BLE radio is intentionally powered off and kept disabled. This prevents radio RF interference from bleeding into the DAC, conserves ~60 KB of critical RAM that the BLE stack would otherwise consume, and eliminates high-priority BLE interrupt latency that causes audio clicks and stutters. All MIDI communication is handled exclusively via class-compliant USB-C MIDI and the 3.5mm TRS hardware MIDI IN jack.
- **Flash Journaling Ring & Dual-Bank OTA Rollback:** Flash persistence uses a compact 24-byte log-structured ring buffer written across a dedicated 4 KB sector (~170 writes before an erase cycle). Flash journaling writes a 24-byte compact record every few minutes to the circular sector ring AND on transport STOP or manual SAVE. Sector erases are scheduled strictly during Eno "velvet-silence" pauses or transport STOP to prevent Execute-In-Place (XIP) flash bus stalls from glitching real-time audio; the next 4 KB journal flash sector is pre-erased during these pauses so the 5% low-battery emergency save executes strictly as a fast 24-byte program write without a blocking erase. Clean flash writes are triggered upon crossing a low-battery threshold (~5% capacity) to prevent flash corruption during sudden voltage drops. For hardware safety, the FM-1 includes a protected bootloader partition (`ota.bin`) with hardware CRC verification that provides dual-bank rollback capabilities (experimental until fully verified on physical hardware).
- **Generative "World" Seeds:** Entire generative session states are represented as a compact 32-bit xorshift PRNG seed combined with macro parameter settings. The full 32-bit xorshift seed is formatted as an 8-character hex code (e.g., `#A3F2-B8D4`), with a 4-character truncated shorthand (e.g., `#D7A4`) used strictly as a display-only shorthand on compact UI views; recall and deterministic session reproduction strictly require the full 32-bit / 8-character hex code.
- **Physical Hardware Knobs & External MIDI Sync:** Dedicated physical knobs provide direct tactile control (Knob 1: Mutation Density, Knob 2: Drone Weight, Knob 3: Shimmer

Speed, Knob 4: Spatial Diffusion/Reverb). The SELECT macro knob ("Weather") smoothly interpolates overall density, BPM (40–75 BPM), drone overtone weight, and thermal portamento drift. When an external MIDI clock signal is detected, the "Weather" macro's internal BPM modulation automatically decouples—keeping the external clock authoritative while continuing to modulate density, drone weight, and portamento.

- **Layer 1 Rhythmic Evolution:** To ensure the 7-step Euclidean drum pattern continuously evolves without feeling repetitive, Layer 1 incorporates per-bar probability gating (chance), dynamic fill muting, and subtle pattern rotations every 2 or 4 bars.
- **Power-of-Two Drop-on-Full FIFO Architecture:** The UI event FIFO buffer is sized to a power of two (32 entries). If the buffer fills during peak rendering, incoming events are dropped automatically, guaranteeing that the audio ISR never blocks or stalls on display operations.
- **Audio I/O Specification:** The 3.5mm analog jack operates as a stereo audio output only (with automatic internal speaker mute). External audio input is supported exclusively via class-compliant USB Audio (UAC 44.1 kHz stereo) over the USB-C connection.

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6. Modular C Firmware Structure (`slambient-fm1`)

The codebase is cleanly modularized for the embedded environment, aligning with the implementation plan across `slambient.h / slambient.c` (Core Generative Co-Processor: scales, random walk, prime clocks, FIFO), `slambient_fx.c` (Master Tape DSP: wow/flutter, saturation, lookahead limiter), and modular hooks into `ui_studio.c`, `audio.c`, and `seq.c`:

- **`slambient.h / slambient.c`**: Core Generative Co-Processor managing scale math, stateful random-walk stepping, prime clocks, modal drift rules, micro-tunings, thermal portamento, voice allocation logic, and lock-free FIFO ring buffer queuing to the UI.
- **`slambient_fx.c`**: Master Tape DSP module handling wow/flutter delay-line modulation, tanh soft-saturation, and lookahead peak limiting on the master output bus.
- **`ui_studio.c, audio.c, seq.c`**: Host system integration points receiving generative events, UI display updates, and master bus audio processing taps.
- **`presets.c / .h`**: Manages onboard storage, recall of custom ambient user slots, instant mood profiles, and Auto-State Snapshots with pre-erased flash journal sector management.

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7. Slambient Hardware UI, Synesthetic Engine & TFT Visualizer

- **Boot Screen:** Custom minimalist "Slambient" startup animation with a breathing analog wave pattern.
- **Synesthetic Color Mapping Engine:** Pitch classes, scale modes, and chord intervals dynamically correspond to specific color coordinates (e.g., deep earthy roots mapping to

warm amber/ochre, ethereal thirds/fifths shifting into twilight indigos, and sparkling arpeggios adding vibrant moss greens or celestial cyans).

- **Harmonic Color Blending:** As notes overlap across the 4 layers, the TFT display real-time blends these chromatic values together across the UI elements, creating an evolving, living watercolor effect that mirrors the audio.
- **TFT Center Display:** Bold, clear text showing the active root note and chord voicing (e.g., `D : WARM_TRIAD`) rendered in the primary synesthetic color of the chord.
- **Harmonic Orbit & Layer Arcs:** A morphing organic center polygon accompanied by four concentric vector arcs representing the 4 prime-number layers as they slowly drift and phase-lock over time.
- **Battery Conservation, Auto-Dim & Diagnostic Instruments:** Includes a 60-second display auto-dimmer that drops backlight brightness to 30% during generative idle, extending battery runtime to 10–12 hours. A development overlay provides real-time diagnostic instruments including an audio buffer underrun counter and a CPU load percentage meter.
- **TFT Display Architecture & Refresh Rate:** Visual refresh rates are locked to a cinematic 24 FPS (filmic / classic hand-drawn animation frame rate) to evoke the vintage Boards of Canada tape aesthetic while capping CPU overhead and SPI bus saturation. Screen updates utilize non-blocking DMA framebuffer transfers and dirty-rectangle rendering for the harmonic orbit polygon, concentric arcs, and watercolor blending to prevent audio buffer underruns.

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8. Hardware Verification & Pre-Flight Checklist

- **Local Build Toolchain:** Verified build pipeline running the JieLi `pi32v2` toolchain (located at `~/.jieli/toolchain`) via Docker Desktop on macOS.
- **Target Binary Package:** Compiles to `build/felucca.fwsc` with package ID `FM-1_900`.
- **Flashing Procedure:** Flashed via USB-C using WebMIDI in Chrome/Edge at fm1.designburgapps.com, featuring instant single-click rollback to stock V15 firmware.
- **Desktop Simulator First & Golden-Seed Regression Tests (Step Zero):** Prior to deploying to physical hardware, compile `generator.c` and `sequencer.c` as a standalone macOS command-line executable outputting MIDI directly to a DAW or via RtMidi. Simulator determinism relies on an abstracted virtual clock with recorded timeline inputs, eliminating unseeded rand() calls or real-time jitter. Automated golden-seed CLI tests verify that identical world seeds produce byte-identical event streams across multi-hour runs before flashing firmware.

9. Development Roadmap & Step-by-Step Implementation Guide

This section outlines the step-by-step development phases focused on the event-driven co-processor architecture, desktop simulator workflow, and recommended build order for robust hardware verification.

Phase 0: Desktop Simulator & Golden-Seed Verification (macOS / RtMidi)

- **Build Command-Line MIDI Simulator:**
  - Compile `generator.c` and `sequencer.c` locally on macOS using clang/gcc linked to RtMidi.
  - Connect MIDI output to a DAW to audit generative event output, prime-number layer drifting, micro-glides, xorshift32 seed reproducibility, and modal shifts. Simulator determinism relies on an abstracted virtual clock with recorded timeline inputs, eliminating unseeded rand() calls or real-time jitter. Execute automated golden-seed CLI tests to ensure byte-identical output across multi-hour sessions.
  - **Phase 0 Statistical Regression Test Suite:**
    - **1. Tension Rule Test:** Asserts that tension interval occurrences stay within 9.0%–11.0% across 1,000,000 steps.
    - **2. Voice Accounting & Leak Test:** Asserts pad voicings never exceed 3 voices, all borrowed voices return, and note-on/note-off counts balance with zero leaks.
    - **3. Window Hash Repetition Audit:** Hashes 64-step event windows across 100,000 steps to verify true non-repetition.

Phase 1: The SLOOP Firmware Hook Audit & Master Bus Tap

- **Master Bus Tap Audit & Pass/Fail Gates:**
  - Audit `audio.c` and `fx.c` to evaluate system integration capabilities. The Phase 1 Hook Audit consists of four explicit Yes/No gate definitions and concrete fallbacks:
    - **Gate 1: Can you tap the post-mix master bus?** If YES: insert custom wow/flutter delay modulation, tanh soft-saturation, and lookahead peak limiter in `slambient_fx.c`. If NO: reuse stock FX loop in `fx.c`.
    - **Gate 2: Can you inject events with sample-accurate sub-block offsets?** If YES: preserve sub-millisecond swing and timing humanization. If NO: accept 1.45 ms block quantization (64-sample audio buffers at 44.1 kHz).
    - **Gate 3: Can you control per-voice or per-track amplitude?** If YES: apply per-voice micro-dynamics and ducking. If NO: apply dynamic ducking via track-level gain.
    - **Gate 4: Can you control the voice allocator?** If YES: implement priority preemption hierarchy. If NO: use track partitioning only (Live Keys on dedicated track).
- **Event Injection Point:**
  - Determine whether SLOOP triggers notes per audio block or supports sample-accurate offsets into the 64-sample audio buffer.
- **Track & Voice Partitioning:**
  - Route layers to dedicated SLOOP tracks (Track 1 = Drone, Track 2 = Chords/Lead & Live Keys, Track 3 = Shimmer Arp, Track 4 = Drums). Live

silicone keys route into Track 2, taking preemption priority over generative chord voices. When Lead and Pad share Track 2, they share the same synth preset timbre so they sound like a unified keyboardist playing melody over chords. Documented confirmation verifies that SLOOP allows up to 6 voices allocated to Track 2 (3 pad + 1 lead + 2 live keys).

- **Voice-Stealing Hierarchy Implementation:**
  - Implement the strict preemption hierarchy (Live Keys > Lead > Pad > Arp > Drone) featuring a 10 ms cosine amplitude ramp-down on voice cuts and preemption.

Phase 2: Embedded Control, Envelope & LFO Engine (`slambient.c / .h`)

- **Fixed-Point Math & Control, Envelope & LFO Engine:**
  - *Coding Primer (Fixed-Point Arithmetic):* To prevent floating-point CPU stalls inside high-priority audio ISRs, all calculations use `q16_t` (Q16.16) integer math where numbers represent 16 integer bits and 16 fractional bits.
  - Implement `generator_init()` and control signal evaluation using the pre-computed `SINE_LUT[1024]` lookup table solely for control LFOs, pitch curves, and tape flutter modulation—not raw audio PCM oscillators.
- **Smooth Voice Amplitude Envelopes:**
  - Implement `generator_release_voice()` with 50–100 ms exponential amplitude release curves for chord and modal transitions, and 10 ms cosine amplitude ramp-downs for voice preemption.

Phase 3: Generative Timing, 6-Prime Clock Dividers & FIFO (`sequencer.c` / `sequencer.h`)

- **6-Prime Multi-Layer Clock Assignments:**
  - Implement independent step counters and clock dividers assigning all six prime cycle lengths: 7-step (Layer 1 Euclidean percussion), 11-step (Layer 3 Shimmer micro-arpeggio), 13-step (Layer 2 Chords and Lead), 17-step (Layer 4 Sub-drone), 19-step (Euclidean pattern rotation & rhythm chance modulation), and 23-step (Macro Weather drift, modal inversion & oblique mutation triggers).
  - *Coding Primer (Modulo Arithmetic):* Wrap counters using modulo arithmetic (`(step + 1) % cycle_length`) to let layers drift naturally over hours without phase collapse.
- **Lock-Free FIFO Integration:**
  - Hook audio trigger events to `ui_fifo_push()` using volatile head and tail pointers for thread-safe lock-free operation between audio ISR and main UI thread.
- **Drum Engine Transient Softening:**
  - Implement `percussion_filter_damp` low-pass filtering on Track 4 drum outputs so Euclidean rhythms sit gently behind pads and arpeggios.

Phase 4: Filmic 24 FPS Display Engine (Heap-allocated DMA dirty rectangles)

- **24 FPS Non-Blocking Frame Limiter:**

    - Enforce an ~41 ms frame period (`UI_FRAME_PERIOD_MS`) in the main loop to achieve a filmic, vintage animation visual aesthetic while capping CPU and SPI bus usage.
- **FIFO Event Processing:**
    - Drain the UI FIFO in `controls_render_frame_24fps()` using `ui_fifo_pop()` to process events generated across each frame window.
- **Heap-Allocated DMA Canvas Tile & Dirty Rectangle Redraws:**
    - Allocate the 4–8 KB canvas tile once at boot from the dynamic audio heap, mark it non-cacheable/DMA-capable with zero runtime alloc/free operations, restrict updates to dirty rectangular regions (center chord text, polygon, arcs), and dispatch updates via non-blocking SPI DMA to prevent audio buffer underruns.

Phase 5: Flash Journaling, Low-Battery Save & 24-Hour Soak / Burn-in Test

- **Compact 24-Byte Circular Journaling & Low-Battery Trigger:**
    - Implement `flash_journal_write_state()` in `presets.c` using the packed 24-byte structure written every few minutes to the circular sector ring, on transport STOP, manual SAVE, or upon reaching a low-battery threshold (~5% capacity). Sector erases are restricted to Eno "velvet-silence" pauses or transport STOP. Validate integrity on boot using the `0x534C414D` magic number and rolling CRC32.
- **Final Compilation & Hardware Flash:**
    - Perform a full build of `build/felucca.fwsc`, verify flash and RAM ceilings, flash to hardware via WebMIDI, and perform a 24-hour soak/burn-in test to guarantee long-term stability and zero audio glitches. Utilize an embedded debug mode dumping the first N events over USB MIDI to diff against the Phase 0 simulator golden file for sim-to-device parity verification.